1. Off centre collision makes the particles go off at an angle (total momentum is conserved)
2.
    a. Initial (before collision): Total momentum is (1.0, 0.0, 0.0), kinetic energy is 2.5, and angular momentum is (0.0, 1.2, -25.0).
    b. Final (after collision): Total momentum is (1.0, 0.0, 0.0), kinetic energy is 2.5, and angular momentum is (0.0, 1.2, -25.0).
    c. Conservation: Momentum, kinetic energy, and angular momentum are all conserved across the elastic collision.
3. Inverse mass is 1/m, so for m=0, inverse mass is infinite, so it behaves like a fixed object since no forces acting on it will affect it.
    a. Wall (v3): Stays at (0, 0, 0)
    b. Ping-pong ball (v1): Reverses to −29.6 in x (about 2.96× the initial speed)
    c. Basketball (v2): Reverses to about −9.6 in x (about 0.96× the initial speed)
    d. Both balls move toad the wall together, at v=10, basketball hits the wall, completely elastic collision and is sent in opposite direction at v=-10, basketball hits ping pong ball and sends it at v=30
    e. The ratios staying the same means the final result is always the same, no matter how they are positioned, however, if you were to change the initial speed, the final velocities scale linearly with v_x. I.e. the final velocity ratios are dependant only on the ratio of masses.
4. Kinetic Energy is conserved, but linear and angular momentum changes, this is because the boundaries of the simulation itself, being walls, act as a means to exchange L and P with the system, but cannot absorb K, hence it is conserved.